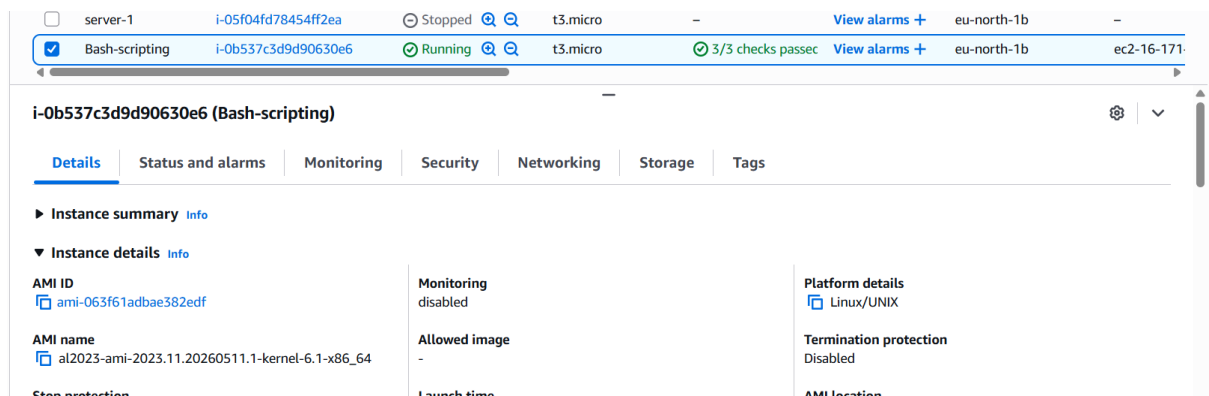


Bash Scripting Video Executions – Day 1

Launch New Instance

Step 1: Launch AWS EC2 Instance

- Open AWS Console
- Go to EC2 Dashboard
- Click **Launch Instance**
- Choose **Amazon Linux**
- Create/Select key pair
- Launch instance



Step 2: Connect to Instance

```
ssh -i mykey.pem ec2-user@Public-IP
```

Move to root:

```
sudo -i
```

Validation:

```
whoami
```

Expected Output:

```
root
```

Screenshot Evidence: Mandatory

[illegible]

Finding Current Shell and Available Shells

Outcome:- Successfully identified current shell, available shells, and user assigned shell in Linux.

Objective:- To learn how to identify shells configured in Linux systems.

Prerequisites:- Linux terminal access

Step-by-Step Implementation

Step 1: Check Current Shell

Command Used

```
echo $0
```

Expected Output -bash

Explanation

\$0 displays currently active shell.

Step 2: Check Available Shells

Command Used cat /etc/shells

Explanation Displays all installed shells available in the system.

Step 3: Check User Assigned Shell

Command Used cat /etc/passwd

Explanation Displays all users and their assigned shells.

Validation Steps

Validation	Expected Result
echo \$0	Shows active shell
cat /etc/shells	Displays installed shells
cat /etc/passwd	Displays user shell assignment

To switch to different shell we can just give sh and check echo \$0 and to come out of shell run exit command.

```
[root@ip-172-31-32-195 scripts]# echo $0
-bash
[root@ip-172-31-32-195 scripts]# cat /etc/shells
/bin/sh
/bin/bash
/usr/bin/sh
/usr/bin/bash
/bin/csh
/bin/tcsh
/usr/bin/csh
/usr/bin/tcsh
[root@ip-172-31-32-195 scripts]# csh
[root@ip-172-31-32-195 scripts]# echp $0
echp: Command not found.
[root@ip-172-31-32-195 scripts]# echo $0
csh
[root@ip-172-31-32-195 scripts]# sh
sh-5.2# exit
exit
[root@ip-172-31-32-195 scripts]# tcsh
[root@ip-172-31-32-195 scripts]# ls
Hello
[root@ip-172-31-32-195 scripts]# ll
total 4
-rwxrwxrwx. 1 root root 28 May 14 09:00 Hello
[root@ip-172-31-32-195 scripts]# cd .
[root@ip-172-31-32-195 scripts]# cd ~
[root@ip-172-31-32-195 ~]# mkdir -p /sasha/scripts
[root@ip-172-31-32-195 ~]# cd /sasha/scrpits/
[root@ip-172-31-32-195 scrpits]# ls
[root@ip-172-31-32-195 scrpits]# pwd
/sasha/scrpits
[root@ip-172-31-32-195 scrpits]# vi HELLOWORLD
[root@ip-172-31-32-195 scrpits]# ls
HELLOWORLD
[root@ip-172-31-32-195 scrpits]# ls 0-ll
ls: cannot access '0-ll': No such file or directory
[root@ip-172-31-32-195 scrpits]# ls -ll
total 4
-rw-r--r--. 1 root root 28 May 14 09:09 HELLOWORLD
[root@ip-172-31-32-195 scrpits]# chmod 777 HELLOWORLD
chmod: cannot access 'HELLOWORLD': No such file or directory
[root@ip-172-31-32-195 scrpits]# chmod 777 HELLOWORLD
[root@ip-172-31-32-195 scrpits]# LS -LL
LS: Command not found.
[root@ip-172-31-32-195 scrpits]# ls -ll
total 4
-rwxrwxrwx. 1 root root 28 May 14 09:09 HELLOWORLD
[root@ip-172-31-32-195 scrpits]#
```

Task 3: Basic System Information Script

Script

```
#!/bin/bash
```

```
pwd
```

```
ls
```

```
whoami
```

```
date
```

```
cal
```

```
touch a b c
```

```
Permission:
```

chmod 755 basic.sh

Execute:

./basic.sh

Validation:

ls

Expected:

```
root@ip-172-31-32-195:/pragati/scripts
[root@ip-172-31-32-195 ~]# mkdir -p /SASHA/scripts
[root@ip-172-31-32-195 ~]# cd /SASHA/scripts
[root@ip-172-31-32-195 scripts]# ls
[root@ip-172-31-32-195 scripts]# pwd
/SASHA/scripts
[root@ip-172-31-32-195 scripts]# vi helloworld
[root@ip-172-31-32-195 scripts]# vi helloworld
[root@ip-172-31-32-195 scripts]# ll
total 4
-rw-r--r--. 1 root root 28 May 14 09:19 helloworld
[root@ip-172-31-32-195 scripts]# chmod 777 helloworld
[root@ip-172-31-32-195 scripts]# ./helloworld
helloworld
[root@ip-172-31-32-195 scripts]# cd
[root@ip-172-31-32-195 ~]# pwd
/root
[root@ip-172-31-32-195 ~]# /SASHA/scripts/helloworld
helloworld
[root@ip-172-31-32-195 ~]#

root@ip-172-31-32-195 ~]# /SASHA/scripts/helloworld
helloworld
[root@ip-172-31-32-195 ~]# ls
[root@ip-172-31-32-195 ~]# cd /SASHA/scripts
[root@ip-172-31-32-195 scripts]# ls
helloworld
[root@ip-172-31-32-195 scripts]# ./helloworld
helloworld
[root@ip-172-31-32-195 scripts]# bash helloworld
helloworld
[root@ip-172-31-32-195 scripts]#
```

Task 1: Create a Basic Shell Script – Hello World

Outcome:- Successfully created and executed a basic shell script that prints “Hello World” as output.

Objective:- The objective of this task is to understand the basic structure of a shell script and learn how to execute commands using Bash scripting.

Prerequisites:- Before starting this task, the following requirements were needed:

- Linux system access
- Bash shell availability
- Basic knowledge of terminal commands
- Text editor such as vi

Step-by-Step Implementation:-

In this task, a basic shell script was created using the Bash shell. First, a new file named helloworld.bash was created using a text editor. Inside the script, the shebang line `#!/bin/bash` was added to define the shell interpreter. Then the echo command was used to print the message “Hello World”.

Script Used

```
#!/bin/bash
echo "Hello World"
```

After saving the file, executable permissions were assigned using the `chmod` command.

Permission Command

```
chmod 755 hello.sh
```

The script was then executed using the following command:

```
./helloworld.bash
```

The output displayed the message “Hello World!” successfully.

```
[root@ip-172-31-32-195 scripts]# cd
[root@ip-172-31-32-195 ~]# ls
[root@ip-172-31-32-195 ~]# cd /SASHA/scripts
[root@ip-172-31-32-195 scripts]# pwd
/SASHA/scripts
[root@ip-172-31-32-195 scripts]# ll
total 4
-rwxrwxrwx. 1 root root 28 May 14 09:19 helloworld
[root@ip-172-31-32-195 scripts]# cat helloworld
#!/bin/bash
echo helloworld
[root@ip-172-31-32-195 scripts]# |
[root@ip-172-31-32-195 scripts]# mv helloworld helloworld.bash
[root@ip-172-31-32-195 scripts]# ll
total 4
-rwxrwxrwx. 1 root root 28 May 14 09:19 helloworld.bash
[root@ip-172-31-32-195 scripts]# |
```

```

-rwxrwxrwx. 1 root root 28 May 14 09:19 helloworld.bash
[root@ip-172-31-32-195 scripts]# vi abcd.bash
[root@ip-172-31-32-195 scripts]# vi abcd.bash
[root@ip-172-31-32-195 scripts]# ll
total 8
-rw-r--r--. 1 root root 22 May 14 09:30 abcd.bash
-rwxrwxrwx. 1 root root 28 May 14 09:19 helloworld.bash
[root@ip-172-31-32-195 scripts]# chmod 777 abcd.bash
[root@ip-172-31-32-195 scripts]# ll
total 8
-rwxrwxrwx. 1 root root 22 May 14 09:30 abcd.bash
-rwxrwxrwx. 1 root root 28 May 14 09:19 helloworld.bash
[root@ip-172-31-32-195 scripts]# chmos 544 abcd.bash
chmos: Command not found.
[root@ip-172-31-32-195 scripts]# chmod 544 abcd.bash
[root@ip-172-31-32-195 scripts]# ll
total 8
-r-xr--r--. 1 root root 22 May 14 09:30 abcd.bash
-rwxrwxrwx. 1 root root 28 May 14 09:19 helloworld.bash
[root@ip-172-31-32-195 scripts]# chmos 544 abcd.bash
chmos: Command not found.
[root@ip-172-31-32-195 scripts]# chmod 544 abcd.bash
[root@ip-172-31-32-195 scripts]# ll
total 8
-r-xr--r--. 1 root root 22 May 14 09:30 abcd.bash
-rwxrwxrwx. 1 root root 28 May 14 09:19 helloworld.bash
[root@ip-172-31-32-195 scripts]# ./abcd.bash
abcd
[root@ip-172-31-32-195 scripts]# bash abcd.bash
abcd
[root@ip-172-31-32-195 scripts]# chmod 644 abcd.bash
[root@ip-172-31-32-195 scripts]# ll
total 8
-rw-r--r--. 1 root root 22 May 14 09:30 abcd.bash
-rwxrwxrwx. 1 root root 28 May 14 09:19 helloworld.bash
[root@ip-172-31-32-195 scripts]# ./abcd.bash
./abcd.bash: Permission denied.
[root@ip-172-31-32-195 scripts]# |
root@ip-172-31-32-195 scripts]# ll
total 8
-rw-r--r--. 1 root root 22 May 14 09:30 abcd.bash
-rwxrwxrwx. 1 root root 28 May 14 09:19 helloworld.bash
root@ip-172-31-32-195 scripts]# ./abcd.bash
./abcd.bash: Permission denied.
root@ip-172-31-32-195 scripts]# chmod 755 abcd.bash
root@ip-172-31-32-195 scripts]# ./abcd.bash
bcd
root@ip-172-31-32-195 scripts]#

```

Task 1: Create a Basic Shell Script – Hello World

Outcome:- Successfully created and executed a basic shell script that prints “Hello World” as output.

Objective:- The objective of this task is to understand the basic structure of a shell script and learn how to execute commands using Bash scripting.

Prerequisites:- Before starting this task, the following requirements were needed:

- Linux system access
- Bash shell availability
- Basic knowledge of terminal commands
- Text editor such as vi

Step-by-Step Implementation:-

In this task, a basic shell script was created using the Bash shell. First, a new file named helloworld.bash was created using a text editor. Inside the script, the shebang line `#!/bin/bash` was added to define the shell interpreter. Then the echo command was used to print the message “Hello World”.

Script Used

```
#!/bin/bash  
echo "Hello World"
```

After saving the file, executable permissions were assigned using the `chmod` command.

Permission Command

```
chmod 755 hello.sh
```

The script was then executed using the following command:

```
./helloworld.bash
```

The output displayed the message “Hello World!” successfully.

Define variables:


```
#!/bin/bash
#Purpose: Script for hello world
#Modification :04/19/2026
#Owner : SASHA
```

```
echo "hello world"
```

```
~
~
~
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~
```


In this task, a shell script was created to execute multiple Linux commands sequentially. A file named basic.sh was created and edited using a text editor. The Bash interpreter line was added at the top of the script.

The script included commands such as:

- pwd to display the present working directory
- ls to list files and directories
- whoami to display current logged-in user
- date to show current date and time
- cal to display calendar
- touch to create empty files

Script Used

```
#!/bin/bash  
pwd  
ls  
whoami  
date  
cal  
touch a b c
```

After saving the script, execute permissions were provided using the chmod command.

```
chmod 755 basic.sh
```

The script was executed using:

```
./basic.sh
```

The output displayed system information and created three empty files named a, b, and c.

Validation Steps:-

Validation was performed by:

- Checking command outputs

Verifying file creation using ls

```
#!/bin/bash
```

```
pwd
```

```
ls
```

```
whoami
```

```
date
```

```
cal
```

```
touch a b c
```

```
~
```

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~
```

```
• :wq!
```

```
commands.bas: Command not found.
```

```
[root@ip-172-31-32-195 scripts]# vi commands.bash
```

```
[root@ip-172-31-32-195 scripts]# ./commands.bash
```

```
./commands.bash: Permission denied.
```

```
[root@ip-172-31-32-195 scripts]# chmod 755 commands.bash
```

```
[root@ip-172-31-32-195 scripts]# ./commands.bash
```

```
/SASHA/scripts
```

```
commands.bas  commands.bash  helloworld.bash
```

```
root
```

```
Thu May 14 10:47:24 UTC 2026
```

```
May 2026
```

```
Su Mo Tu We Th Fr Sa
```

```
1 2
```

```
3 4 5 6 7 8 9
```

```
10 11 12 13 14 15 16
```

```
17 18 19 20 21 22 23
```

```
24 25 26 27 28 29 30
```

```
31
```

```
[root@ip-172-31-32-195 scripts]# |
```

```
[root@ip-172-31-32-195 scripts]# date
Thu May 14 10:49:57 UTC 2026
[root@ip-172-31-32-195 scripts]# cal
      May 2026
Su Mo Tu We Th Fr Sa
                1  2
 3  4  5  6  7  8  9
10 11 12 13 14 15 16
17 18 19 20 21 22 23
24 25 26 27 28 29 30
31
[root@ip-172-31-32-195 scripts]#
```

```
[root@ip-172-31-32-195 scripts]# pwd
```

Task 3: Perform Basic

Administration Tasks Using Linux Commands

Outcome:- Successfully executed Linux administration commands to monitor system performance and resource utilization.

Objective:- The objective of this task is to learn commonly used Linux administration commands for monitoring CPU, memory, disk usage, uptime, and system performance.

Step-by-Step Implementation:-

In this task, multiple Linux administration commands were executed individually from the terminal to monitor server resources and system health.

The following commands were used:

Commands Used

top
df -h
free -m
uptime
iostat

The top command continuously monitors running processes and resource utilization. Since it continuously refreshes output and appears stuck, the following command was used to limit the output:

top | head -10

The df -h command displayed disk usage in human-readable format, while free -m displayed memory usage in megabytes.

The uptime command displayed server running time and load averages. The iostat command was used to monitor CPU and disk statistics.

Validation Steps:-

- Validation was performed by confirming:
- Resource statistics displayed correctly

Disk and memory usage information appeared

[illegible]

```
/root/scripts
[root@ip-172-31-32-195 scripts]# vi admin.bash
[root@ip-172-31-32-195 scripts]# ls
a admin.bash b c commands.bash commands.bash helloworld.bash
[root@ip-172-31-32-195 scripts]# ll
total 16
-rw-r--r-- 1 root root 0 May 14 10:47 a
-rw-r--r-- 1 root root 49 May 14 10:46 admin.bash
-rw-r--r-- 1 root root 0 May 14 10:47 b
-rw-r--r-- 1 root root 0 May 14 10:47 c
-rw-r--r-- 1 root root 48 May 14 10:49 commands.bash
-rwxr-xr-x 1 root root 49 May 14 10:46 commands.bash
-rwxr-xr-x 1 root root 118 May 14 10:33 helloworld.bash
[root@ip-172-31-32-195 scripts]# chmod 755 admin.bash
[root@ip-172-31-32-195 scripts]# ./admin.bash
top - 11:00:41 up 3:20, 3 users, load average: 0.00, 0.00, 0.00
Tasks: 111 total, 1 running, 110 sleeping, 0 stopped, 0 zombie
%cpu(s): 0.2 us, 0.0 sy, 0.0 ni, 99.8 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 916.8 total, 535.3 free, 140.8 used, 240.7 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 639.4 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR  S  %CPU  %MEM     TIME+ COMMAND
    1 root        20   0 173244 17644 10984 S   0.0   1.9   0:01.47 systemd
    2 root        20   0      0      0      0 S   0.0   0.0   0:00.00 kthreadd
    3 root        20  -20      0      0      0 I   0.0   0.0   0:00.00 rcu_gp
    4 root        20  -20      0      0      0 I   0.0   0.0   0:00.00 rcu_par_gp
    5 root        20   0      0      0      0 I   0.0   0.0   0:00.00 slab_flushdq
    6 root        20  -20      0      0      0 I   0.0   0.0   0:00.00 netns
    8 root        20  -20      0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
   10 root        20  -20      0      0      0 I   0.0   0.0   0:00.00 mpagecpusq
   11 root        20   0      0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
   12 root        20   0      0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
   13 root        20   0      0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
   14 root        20   0      0      0      0 S   0.0   0.0   0:00.09 ksoftirqd/0
   15 root        20   0      0      0      0 I   0.0   0.0   0:00.16 rcu_preempt
   16 root        20   0      0      0      0 S   0.0   0.0   0:00.01 migration/0
   18 root        20   0      0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
   19 root        20   0      0      0      0 S   0.0   0.0   0:00.00 cpuhp/1
   20 root        20   0      0      0      0 S   0.0   0.0   0:00.07 migration/1
   21 root        20   0      0      0      0 S   0.0   0.0   0:00.10 ksoftirqd/1
   23 root        20  -20      0      0      0 I   0.0   0.0   0:00.00 kworker/1:0H-events_highpri
   26 root        20   0      0      0      0 S   0.0   0.0   0:00.00 kdevtmpfs
   27 root        20  -20      0      0      0 I   0.0   0.0   0:00.00 inet_frag_wq
   28 root        20   0      0      0      0 S   0.0   0.0   0:00.00 kauditd
   29 root        20   0      0      0      0 S   0.0   0.0   0:00.00 khungtaskd
   30 root        20   0      0      0      0 S   0.0   0.0   0:00.00 koom_reaper
   32 root        20  -20      0      0      0 I   0.0   0.0   0:00.00 writeback
   33 root        20   0      0      0      0 S   0.0   0.0   0:00.31 kcompactd0
   34 root        20  -20      0      0      0 S   0.0   0.0   0:00.00 khugepaged
   35 root        20  -20      0      0      0 I   0.0   0.0   0:00.00 cryptd
   36 root        20  -20      0      0      0 I   0.0   0.0   0:00.00 kintegrityd
   37 root        20  -20      0      0      0 I   0.0   0.0   0:00.00 kblockd
   38 root        20  -20      0      0      0 I   0.0   0.0   0:00.00 blkcg_punt_bio
   40 root        20  -20      0      0      0 I   0.0   0.0   0:00.00 tpm_dev_wq
   41 root        20  -20      0      0      0 I   0.0   0.0   0:00.00 ed
   42 root        20  -20      0      0      0 I   0.0   0.0   0:00.00 edac-poller
   43 root        20  -20      0      0      0 S   0.0   0.0   0:00.00 watchdogd
   44 root        20  -20      0      0      0 I   0.0   0.0   0:00.01 kworker/0:1H-kblockd
   67 root        20   0      0      0      0 S   0.0   0.0   0:00.00 kswapd0
   70 root        20  -20      0      0      0 I   0.0   0.0   0:00.00 xfsailloc
   71 root        20  -20      0      0      0 I   0.0   0.0   0:00.00 xfs_mru_cache
   74 root        20  -20      0      0      0 I   0.0   0.0   0:00.00 kthrotld
  122 root        20  -20      0      0      0 I   0.0   0.0   0:00.00 nvme-wq
  124 root        20  -20      0      0      0 I   0.0   0.0   0:00.00 nvme-reset-wq
  125 root        20  -20      0      0      0 S   0.0   0.0   0:00.00 cpupl/1

Filesystem      Size  Used Avail Use% Mounted on
devtmpfs         4.0M    0  4.0M   0% /dev
tmpfs            459M    0  459M   0% /dev/shm
tmpfs           184M  440K  183M   1% /run
/dev/nvme0n1p1   8.0G  1.6G  6.4G  20% /
tmpfs            459M    0  459M   0% /tmp
/dev/nvme0n1p128 10M  1.3M  8.7M  13% /boot/efi
tmpfs            92M    0   92M   0% /run/user/1000
               total    used    free   shared  buff/cache   available
Mem:           916        141        534           0        240        638
Swap:            0           0           0

  11:13:32 up 3:33, 3 users, load average: 0.00, 0.00, 0.00
Linux 6.1.170-210.320.amzn2023.x86_64 (ip-172-31-32-195.eu-north-1.compute.inter
nal)    05/14/26      _x86_64_      (2 CPU)

avg-cpu:  %user   %nice %system %iowait  %steal   %idle
           0.07    0.00    0.04    0.01    0.01   99.87

Device            tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_w
rtn
nvme0n1           1.01      12.05      42.80         0.00     154403     548
496              0
```

When we use head-10

```
[root@ip-172-31-32-195 scripts]# vi admin.bash
[root@ip-172-31-32-195 scripts]# ./admin.bash
top - 11:16:53 up 3:36, 3 users, load average: 0.00, 0.00, 0.00
Tasks: 110 total, 1 running, 109 sleeping, 0 stopped, 0 zombie
%cpu(s): 3.2 us, 0.0 sy, 0.0 ni, 96.8 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 916.8 total, 530.9 free, 144.8 used, 241.1 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 635.4 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR  S  %CPU  %MEM     TIME+ COMMAND
    1 root        20   0 173244 17644 10984 S   0.0   1.9   0:01.52 systemd
    2 root        20   0      0      0      0 S   0.0   0.0   0:00.00 kthreadd
    3 root        20  -20      0      0      0 I   0.0   0.0   0:00.00 rcu_gp
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs         4.0M    0  4.0M   0% /dev
tmpfs            459M    0  459M   0% /dev/shm
tmpfs           184M  440K  183M   1% /run
/dev/nvme0n1p1   8.0G  1.6G  6.4G  20% /
tmpfs            459M    0  459M   0% /tmp
/dev/nvme0n1p128 10M  1.3M  8.7M  13% /boot/efi
tmpfs            92M    0   92M   0% /run/user/1000
               total    used    free   shared  buff/cache   available
Mem:           916        144        530           0        241        635
Swap:            0           0           0

  11:16:56 up 3:37, 3 users, load average: 0.00, 0.00, 0.00
Linux 6.1.170-210.320.amzn2023.x86_64 (ip-172-31-32-195.eu-north-1.compute.internal)    05/14/26      _x86_64_      (2 CPU)

avg-cpu:  %user   %nice %system %iowait  %steal   %idle
           0.07    0.00    0.04    0.01    0.01   99.87

Device            tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd
nvme0n1           1.00      11.86      42.75         0.00     154443     556668         0

[root@ip-172-31-32-195 scripts]#
```

- **Task 4: Defining Variables in Bash Script**

Outcome:- Successfully created and executed a shell script using variables.

Objective:- The objective of this task is to understand how variables are declared and used in Bash scripting.

Step-by-Step Implementation:-

In this task, variables were defined inside a shell script to store Linux commands. A script file was created and variables were assigned values such as pwd, ls, whoami, date, and cal.

The variables were later executed using the \$ symbol.

The script was saved and executed after assigning execute permissions.

Validation Steps:- Validation was performed by checking whether all commands stored in variables executed successfully.

Explanation

Variables are used to store values temporarily inside scripts. In Bash scripting, variables can store strings, commands, numbers, or user input. The \$ symbol is used to access variable values.

```
#!/bin/bash
#Define Variables
p=pwd
l=ls
wi=whoami
d=date
c='cal 2026'

echo Run variables tasks
echo
$p
$l
$wi
$d
$c|
~
~
~
~
~
~
~
~
~
~
-- REPLACE --
```



```
Linux 6.1.170-210.320.amzn2023.x86_64 (ip-172-31-32-195.eu-north-1.compute.internal) 05/14/26 _x86_64_ (2 CPU)
[root@ip-172-31-32-195 scripts]# vi variables.bash
[root@ip-172-31-32-195 scripts]# ./variables.bash
./variables.bash: Permission denied.
[root@ip-172-31-32-195 scripts]# chmod 755 variables.bash
[root@ip-172-31-32-195 scripts]# ./variables.bash
Run variables tasks

/SASHA/scripts
a b commands.bas helloworld.bash
admin.bash c commands.bash variables.bash
root
Thu May 14 11:28:11 UTC 2026

2026

January February March
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
 4 5 6 7 8 9 10 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21
11 12 13 14 15 16 17 15 16 17 18 19 20 21 15 16 17 18 19 20 21
18 19 20 21 22 23 24 22 23 24 25 26 27 28 22 23 24 25 26 27 28
25 26 27 28 29 30 31 29 30 31

April May June
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
 5 6 7 8 9 10 11 1 2 3 4 5 6 1 2 3 4 5 6
12 13 14 15 16 17 18 3 4 5 6 7 8 9 7 8 9 10 11 12 13
19 20 21 22 23 24 25 10 11 12 13 14 15 16 14 15 16 17 18 19 20
26 27 28 29 30 17 18 19 20 21 22 23 21 22 23 24 25 26 27
28 29 30 24 25 26 27 28 29 30 28 29 30
31

July August September
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
 5 6 7 8 9 10 11 1 2 3 4 5 1 2 3 4 5
12 13 14 15 16 17 18 2 3 4 5 6 7 8 6 7 8 9 10 11 12
19 20 21 22 23 24 25 9 10 11 12 13 14 15 13 14 15 16 17 18 19
26 27 28 29 30 31 16 17 18 19 20 21 22 20 21 22 23 24 25 26
28 29 30 23 24 25 26 27 28 29 27 28 29 30
30 31

October November December
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
 4 5 6 7 8 9 10 1 2 3 4 5 6 7 1 2 3 4 5
11 12 13 14 15 16 17 8 9 10 11 12 13 14 6 7 8 9 10 11 12
18 19 20 21 22 23 24 15 16 17 18 19 20 21 13 14 15 16 17 18 19
25 26 27 28 29 30 31 22 23 24 25 26 27 28 20 21 22 23 24 25 26
29 30 27 28 29 30 31

[root@ip-172-31-32-195 scripts]#
[root@ip-172-31-32-195 scripts]# cat variables.bash
#!/bin/bash
#Define Variables
p=pwd
l=ls
wi=whoami
d=date
c='cal 2026'

echo Run variables tasks
echo
$p
$l
$wi
$d
$c
[root@ip-172-31-32-195 scripts]#
```

Input and ouput script:

Ex1:

```
#!/bin/bash
```

```
echo Hello, my name is Sasha
```

```
echo
```

```
echo what is your name?
```

echo

~~~~~

echo Hello, my name is \$a

echo

echo what is your name?

read b

echo

echo Hello \$b

echo **Task 5: Input and Output Shell Script Using read and echo**

**Outcome:-** Successfully created interactive shell scripts that accept user input and display dynamic output.

**Objective:-** The objective of this task is to understand how to interact with users in shell scripting using read and echo commands.

**Step-by-Step Implementation:-**

In this task, interactive shell scripts were created using read and echo. The echo command was used to print output on the terminal, while the read command accepted user input dynamically.

The first script asked the user to enter their name and displayed a greeting message.

Both scripts were executed successfully after assigning execute permissions.

**Validation Steps:-** Validation was completed by:

Entering user input successfully

Verifying dynamic greeting output

Checking hostname display

**Explanation**

The read command pauses script execution and waits for user input. The echo command displays output messages. Backticks ( `` ) are used for command substitution, meaning command output is stored inside a variable.

**Common Errors and Troubleshooting**

| Error                   | Cause                      | Resolution              |
|-------------------------|----------------------------|-------------------------|
| Input not accepted      | Missing read command       | Add read statement      |
| Variable empty          | Incorrect variable usage   | Use \$variable          |
| Hostname not displaying | Command substitution issue | Use backticks correctly |

```
#!/bin/bash
```

```
#we need to use the ticks if you want to use this in echo or else not required
```

```
a=hostname
```

```
echo Hello, my name is $a
```

```
echo
```

```
echo what is your name?
```

```
read b
```

```
echo
```

```
echo Hello $b
```

```
echo
```

```
~
```

```
~
```

```
~
```

```
~
```

```
~
```

```
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```

```
~
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```
~
```

```
~
```

```
~
```

```
~
```

```
~
```

```
[root@ip-172-31-32-195 scripts]# vi output.sh
```

```
[root@ip-172-31-32-195 scripts]# ls
```

```
a          b  commands.bas  helloworld.bash  variables.bash
```

```
admin.bash c  commands.bash input.bash
```

```
[root@ip-172-31-32-195 scripts]# vi variables.bash
```

```
[root@ip-172-31-32-195 scripts]# vi output.sh
```

```
[root@ip-172-31-32-195 scripts]# chmod 755 output.sh
```

```
[root@ip-172-31-32-195 scripts]# ./output.sh
```

```
Hello, my name is hostname
```

```
what is your name?
```

```
Pragathi
```

```
Hello Pragathi
```

```
[root@ip-172-31-32-195 scripts]#
```

[illegible]

```
[root@ip-172-31-32-195 scripts]# hostname
ip-172-31-32-195.eu-north-1.compute.internal
[root@ip-172-31-32-195 scripts]# vi output.sh
[root@ip-172-31-32-195 scripts]# ./output.sh
./output.sh: Command not found.
[root@ip-172-31-32-195 scripts]# ./output.sh
Hello, my name is ip-172-31-32-195.eu-north-1.compute.internal

what is your name?
paragthi

Hello paragthi

[root@ip-172-31-32-195 scripts]# hostname
ip-172-31-32-195.eu-north-1.compute.internal
[root@ip-172-31-32-195 scripts]#
```